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JAVA PROJECT REPORT ON

INSURANCE POLICY MANAGEMENT AND PREMIUM REMAINDER SYSTEM

Submitted by

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Semester: **II**

DECLARATION

I hereby declare that the project entitled "**INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **MUGILAN.R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(JAYADHANUSHIKA.S.I)

CERTIFICATE

This is to certify that the project entitled **"INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM"** is a **bona fide work** carried out by **JAYADHANUSHIKA.S.I B.Sc. IT 'B' (2025-2028)**, **INFORMATION TECHNOLOGY ,KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming.**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore
Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The ***Insurance Policy Management and Premium Reminder System*** is a Java-based application developed to simplify and automate the management of insurance policies and premium payments. In traditional systems, policy management is often handled manually, which leads to inefficiencies such as missed premium deadlines, poor record management, and increased workload for employees.

This project aims to provide a computerized solution that helps insurance companies and agents manage customer policy details effectively. The system stores customer information, policy details, premium amounts, due dates, and payment history in an organized manner. One of the most important features of the system is the automatic premium reminder, which alerts users about upcoming due dates to avoid policy lapses.

The application is developed using Core Java concepts such as object-oriented programming, classes, objects, methods, loops, conditional statements, and exception handling. The system can be implemented as a console-based application or extended into a graphical user interface using Java Swing.

The project improves efficiency by reducing manual errors, saving time, and ensuring timely payment of premiums. It also enhances customer satisfaction by providing reminders and maintaining accurate records. This system can be further upgraded with database integration and SMS/email notification features.

1.INTRODUCTION

In today's digital world, automation plays a crucial role in improving efficiency and accuracy in business operations. The insurance industry handles a large volume of customer data, policy details, and premium transactions. Managing these records manually is not only time-consuming but also prone to errors and mismanagement.

The Insurance Policy Management and Premium Reminder System is designed to address these challenges by providing a digital platform for managing insurance-related activities. The system allows users to store customer details, manage policies, track premium payments, and generate reminders for upcoming due dates.

Manual systems often result in missed premium payments, leading to policy lapses and customer dissatisfaction. This system ensures that reminders are generated automatically, reducing the chances of missed deadlines. It also helps insurance agents maintain organized records and provide better service to their clients.

The project is developed using Java programming language and demonstrates the practical application of object-oriented programming concepts. It is a simple yet effective solution that can be used in small insurance agencies or as a learning project for students.

1.1 Objective

The main objective of this project is to develop an efficient system for managing insurance policies and premium payments.

Specific Objectives:

- To design a system that stores customer and policy details systematically
- To automate premium calculation and payment tracking
- To provide reminders for upcoming premium due dates
- To reduce manual errors and improve accuracy
- To develop a user-friendly application using Java
- To demonstrate object-oriented programming concepts
- To improve efficiency in insurance management

Additionally, the system aims to enhance customer satisfaction by ensuring timely notifications and proper record maintenance.

1.2 Overview

The system is a menu-driven Java application that allows users to manage insurance policies effectively. It provides functionalities such as adding new policies, viewing policy details, updating records, and generating reminders for premium payments.

Users can enter customer details such as name, policy number, premium amount, and due date. The system stores this information and processes it whenever required. One of the key features is the reminder system, which alerts users before the premium due date.

The application simplifies the entire process of policy management and ensures that all records are maintained accurately. It can be used as a standalone system or integrated with a database for larger applications.

1.3 Features

The *Insurance Policy Management and Premium Reminder System* includes several important features that make it efficient and easy to use:

- **Customer Management**
  The system allows users to store and manage customer details such as name, policy number, and contact information.
- **Policy Management**
  It helps in creating and maintaining insurance policy records, including policy type, premium amount, and duration.
- **Premium Tracking**
  The system keeps track of premium payments and due dates, ensuring accurate monitoring of payment status.
- **Automatic Reminder System**
  One of the key features is the reminder functionality, which alerts users about upcoming or overdue premium payments.
- **Data Storage and Retrieval**
  The system stores data in an organized manner and allows users to retrieve information quickly whenever needed.
- **User-Friendly Interface**
  The application is designed with a simple menu-driven interface, making it easy to use for beginners.

- **Error Reduction**
  By automating calculations and data handling, the system reduces the chances of manual errors.
- **Time Efficiency**
  The system saves time by performing operations quickly and efficiently compared to manual methods

2.PROGRAM ANALYSIS

Program analysis is a crucial phase in the software development life cycle. It involves studying the existing system, identifying its limitations, understanding user requirements, and proposing an efficient solution. This phase helps in clearly defining what the system should do and how it should perform its functions.

In the context of the ***Insurance Policy Management and Premium Reminder System***, program analysis plays an important role in identifying the challenges faced in traditional insurance management systems and designing a solution that improves efficiency, accuracy, and reliability.

The analysis process includes understanding the current manual system, identifying the problems associated with it, and determining the functional and non-functional requirements of the proposed system.

2.1 Problem Definition

In many insurance companies, policy management is often handled manually using paper records or simple tools. This method creates several problems such as difficulty in maintaining accurate records, delays in retrieving information, and increased chances of human error.

One of the major issues is the lack of a proper reminder system. Customers may forget to pay their premiums on time, which can lead to policy lapses and financial loss. Manual tracking of due dates is time-consuming and unreliable.

Other common problems include:

- Errors in data entry and calculations
- Difficulty in managing multiple policies
- Poor record organization
- Increased workload for employees

Therefore, there is a need for an automated system that can efficiently manage policy details and provide timely premium reminders. The proposed system helps to solve these issues by introducing a computerized solution.

2.2 Project Overview

The *Insurance Policy Management and Premium Reminder System* is a Java-based application designed to simplify insurance operations. It allows users to store customer details, manage policy information, and track premium payments efficiently.

The system enables users to add new policies, update records, and view policy details whenever required. It also includes a reminder feature that alerts users about upcoming premium due dates.

The main functions of the system include:

- Managing customer and policy details
- Tracking premium payments
- Generating reminders for due dates
- Displaying policy records

The system is developed using Core Java concepts and can be implemented as a console-based or GUI-based application. It helps reduce manual effort and improves accuracy.

2.3 Scopes and Limitations

Scope

The system is designed to manage basic insurance operations efficiently. It includes storing customer information, managing policy details, tracking premium payments, and generating reminders.

The system can be used in:

- Insurance companies
- Agencies
- Small business organizations

It helps improve efficiency, reduce errors, and maintain proper records.

Limitations

- The system is basic and may not support large-scale data without database integration
- It does not include advanced features like SMS or email notifications
- Data must be entered manually
- Limited security features

These limitations can be improved in future versions by adding database support and advanced technologies.

3.SYSTEM DESIGN

System design is an important stage in software development where the overall structure of the system is planned before implementation. It defines how the system will work, how different components are organized, and how data flows within the application. A good system design improves efficiency, reduces complexity, and makes the system easy to maintain.
The Insurance Policy Management and Premium Reminder System is designed as a menu-driven Java application. The system allows users to manage insurance policies, track premium payments, and generate reminders. The design follows a modular approach, where the system is divided into smaller parts called modules. Each module performs a specific function, making the system more organized and easier to develop.

3.1.System Architecture

The system consists of three main parts:

- Input – Accepts user details such as customer name, policy number, premium amount, and due date
- Processing – Performs operations like storing data, calculating premiums, and checking due dates
- Output – Displays policy details and reminder messages

This simple structure ensures smooth working of the system.

3.2.Modules of the System

The system is divided into the following modules:

1. Customer Module

Stores and manages customer details such as name and policy number. It allows adding and viewing customer records.

2. Policy Module

Handles policy-related information including policy type, duration, and premium amount. It helps in organizing policy data efficiently.

3. Premium Module

Calculates and tracks premium payments. It ensures accurate calculation and updates payment status.

4. Reminder Module

Generates alerts for upcoming or overdue premium payments. It compares the current data with the due date and displays reminder messages.

5. Report Module

Displays customer and policy details in a clear format. It helps users quickly access stored information.

6. Exit Module

Allows the user to safely exit the program.

3.3.Data Flow

The system works in the following steps:

1. User enters customer and policy details
2. Data is stored in the system
3. Premium details are processed
4. Due dates are checked
5. Reminder is generated if required
6. Output is displayed

4.METHODOLOGY

Methodology refers to the systematic approach followed to develop the project. It explains the step-by-step process used from understanding the problem to designing, implementing, testing, and documenting the system. A proper methodology ensures that the project is completed in an organized and efficient manner.

For the ***Insurance Policy Management and Premium Reminder System***, a structured development process was followed to ensure that all requirements are met and the system works correctly. The development process consists of several important stages, each contributing to the successful completion of the project.

The main stages involved in the methodology are:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the existing system is studied carefully to understand its limitations and issues. The problems in manual insurance management systems are identified, such as difficulty in maintaining records, lack of reminder systems, and chances of human error.

The requirements of the new system are clearly defined based on these problems. The need for an automated system that can manage policy details and provide premium reminders is identified. This phase helps in understanding what features should be included in the system.

4.2. System Design

After analyzing the problem, the next step is to design the system. In this phase, the structure of the system is planned, and different modules are identified.

The system is divided into modules such as:

- Customer Module
- Policy Module
- Premium Module
- Reminder Module
- Report Module

The overall workflow of the system is also designed to ensure smooth functioning. Proper system design helps in reducing errors during development and makes the system easy to understand.

4.3. Program Development

In this phase, the actual coding of the system is done using the Java programming language. The design created in the previous phase is converted into a working program.

The following steps are involved:

- Creating classes and objects
- Writing methods for different operations
- Implementing menu-driven logic
- Handling user input
- Writing code for premium calculation and reminders

Core Java concepts such as loops, conditional statements, and exception handling are used during development. This phase is important because it transforms the design into a functional application.

4.4. Testing and Validation

s.no	Test Case	Input Given	Expected Result	Actual Result
1	Enter Customer Name	Ravi	Name stored correctly	Name stored correctly
2	Enter Policy Number	POL101	Policy number accepted	Accepted
3	Enter Premium Amount	5000	Premium stored correctly	Stored correctly

4	Enter Due Days (Normal)	10	Message: Not due soon	Displayed correctly
5	Enter Due Days (Reminder Case)	3	Reminder message displayed	Displayed correctly
6	Enter Due Days (Overdue Case)	-2	Overdue warning message displayed	Displayed correctly
7	Invalid Input (Text for number)	"abc"	Error or invalid input handled	Error shown
8	Display Policy Details	Stored data	All details displayed correctly	Displayed correctly

4.5. Result and Documentation

After successful testing, the results of the system are analyzed and documented. The system is able to store policy details, track premium payments, and generate reminders effectively.

Proper documentation is prepared to explain how the system works, including its design, implementation, and output. This helps in understanding the project clearly and is useful for future reference.

4.6. Algorithm

The algorithm represents the logical steps followed by the system:

1. Start the program
2. Enter customer details
3. Add policy information
4. Store premium amount and due date
5. Check current date with due date
6. If due date is near, generate reminder
7. Display policy details

8. End the program

4.7. Flowchart

The flowchart shows the graphical representation of the system process.

Basic flow:

Start
↓
Enter Details
↓
Store Data
↓
Check Due Date
↓
Generate Reminder
↓
Display Output
↓
End

5.SOURCE CODE

```
import java.util.Scanner;


public class InsuranceSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String name, policyNo;
        double premium;
        int dueDays;

        System.out.println("=== INSURANCE POLICY SYSTEM ===");

        // Input details
```

```
        System.out.print("Enter Customer Name: ");
        name = sc.nextLine();

        System.out.print("Enter Policy Number: ");
        policyNo = sc.nextLine();

        System.out.print("Enter Premium Amount: ");
        premium = sc.nextDouble();

        System.out.print("Enter Days Left for Due Date: ");
        dueDays = sc.nextInt();

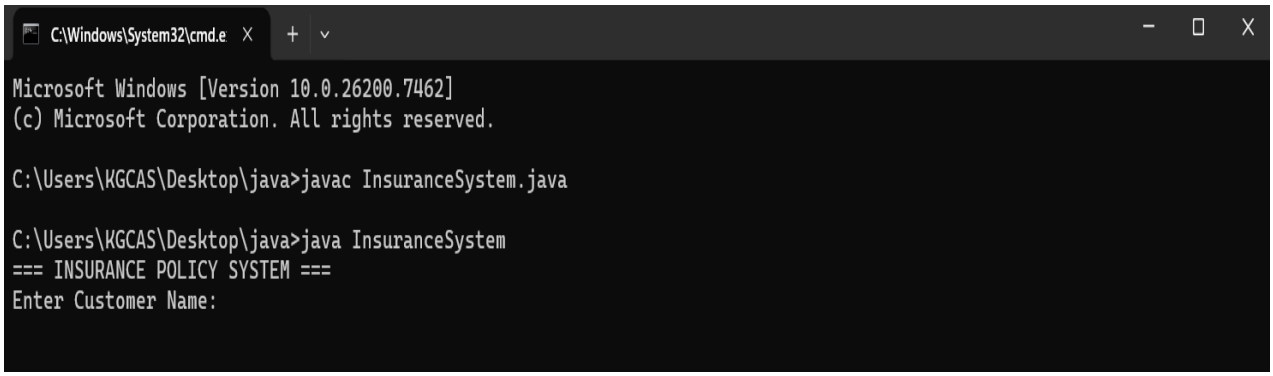
        // Display details
        System.out.println("\n--- POLICY DETAILS ---");
        System.out.println("Customer Name : " + name);
        System.out.println("Policy Number : " + policyNo);
        System.out.println("Premium Amount: " + premium);

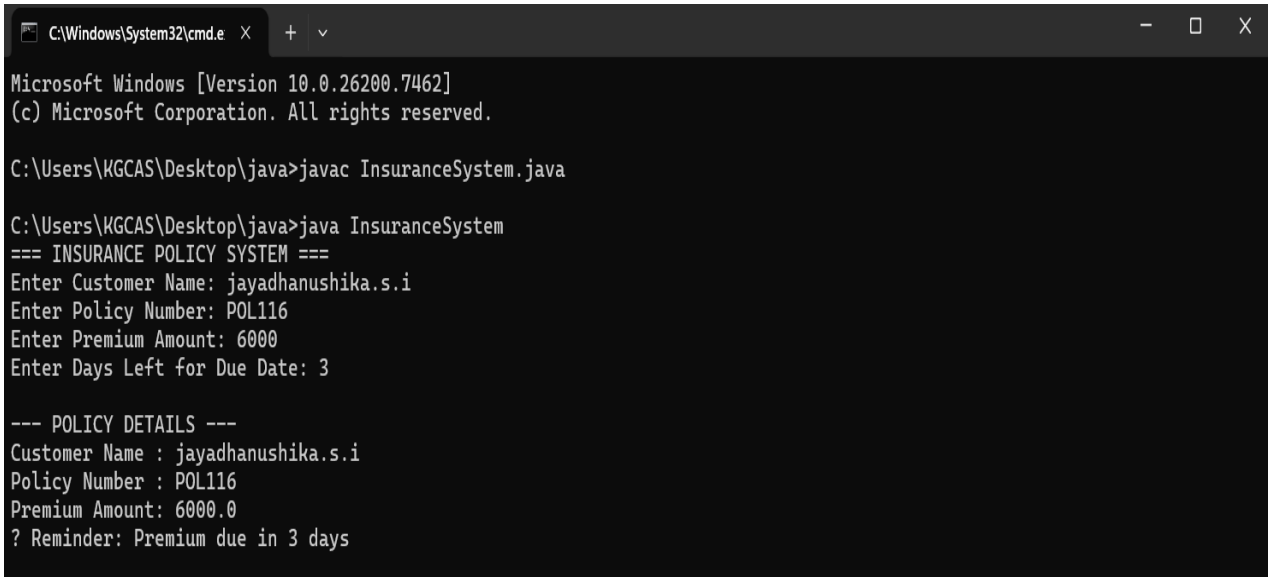
        // Reminder logic
        if (dueDays < 0) {
            System.out.println("⚠ Premium is OVERDUE!");
        } else if (dueDays <= 5) {
            System.out.println("□ Reminder: Premium due in " + dueDays + " days");
        } else {
             System.out.println("✓ Premium payment is not due soon");
        }

        sc.close();
    }
}
```

6.INPUT AND OUTPUT SCREENS







7.DISCUSSION & CONCLUSION

7.1 Discussion

The ***Insurance Policy Management and Premium Reminder System*** was developed to simplify the process of managing insurance policies and tracking premium payments. The system helps overcome the problems of manual record keeping, such as errors, delays, and lack of proper reminders.

During the development of this project, Core Java concepts like classes, objects, methods, and conditional statements were used. The system successfully stores policy details, processes premium information, and generates reminders based on due dates.

The reminder feature is one of the most important parts of the system, as it helps users avoid missing premium payments. This improves efficiency and reduces the chances of policy lapses.

The system was tested with different inputs, and it produced accurate results. The menu-driven design also makes it easy to use for users with basic knowledge.

However, the system is a basic version and does not include advanced features like database storage or automatic notifications. These can be added in future improvements.

7.2 Conclusion

The ***Insurance Policy Management and Premium Reminder System*** is a simple and effective application developed using Java. It helps in managing policy details, tracking premium payments, and generating reminders.

The project achieves its main objective of reducing manual work and improving accuracy. It provides a structured way to store and manage insurance data.

This project also helps in understanding how programming concepts can be applied to real-world problems. It demonstrates the importance of automation in improving efficiency.

In the future, the system can be enhanced by adding features such as database integration, graphical user interface, and notification systems.

Overall, the project is useful, practical, and provides a good foundation for developing more advanced applications.

8.REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*
• E. Balagurusamy – *Programming with Java*

Websites

http://www.oracle.com/java
http://www.geeksforgeeks.org
http://www.javatpoint.com
http://www.w3schools.com/java